Domain Invariant and Compact Prototype Contrast Adaptation for Hyperspectral Image Classification

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***Abstract*— Contrastive learning achieves good performance on hyperspectral image classification (HSIC), but its application on cross-scene classification is still challenging due to domain shifts. The emergence of domain adaptation (DA) techniques can reduce domain discrepancy and transfer a model between two domains. Recently, instance-level contrast adaptation methods can connect two related domains, and domain-invariant features are extracted. However, it is sensitive to noisy samples and only learns low-level discriminative features. To solve these problems, a novel domain invariant and compact prototype contrast adaptation (DIC-proCA) framework is proposed for HSIC. About the proposed DIC-proCA, the prototype is introduced into the contrastive learning framework, which serves as a representative embedding of semantically similar samples, has class representativeness, and can alleviate the negative impact of outliers. Taking into account the class representativeness of the prototype and the discriminability of the sample itself, a bidirectional interdomain instance-to-prototype contrastive loss is proposed. It explicitly expresses feature relationships between categories in different domains, and then extracts domain-invariant features. Meanwhile, the mining of compact discriminative features within the target domain is facilitated by instance-level contrastive learning after data augmentation. In addition, the strategy of label smoothing (LS) promotes the clusters in the domain to be more compact and evenly separated, making the model more generalizable. Three cross-scene HSIC tasks demonstrate that the proposed DIC-proCA exhibits superior performance compared to some advanced DA algorithms.**

***Index Terms*— Contrastive learning, hyperspectral image classification (HSIC), prototype contrast adaptation.**

I. Introduction

THE rapid development of remote sensing (RS) imaging technology allows different types of RS data to be obtained for many applications such as disaster monitoring and mineral exploration [1], [2], [3], [4], [5]. Due to the rich spectral information contained in hundreds of bands, the subtle spectral characteristics of different substances can be reflected in a hyperspectral image (HSI) for fine classification. Hence, HSI classification (HSIC) is a common and popular topic, which aims to label every pixel in the scene [6], [7], [8], [9]. Most existing classification methods may require more labeled samples to achieve good classification performance. However, due to the complex terrain environment, labeling HSI is a large-scale project that requires a significant amount of time and labor costs [10], [11], [12]. A straightforward idea is to directly apply a model trained on one dataset (source domain) to a new dataset (target domain), but this cannot produce satisfactory results since in real situations, the two datasets are usually not independent and identically distributed. Moreover, due to the effects of different sensors, collection conditions, and so on, the same substance may not have similar spectral characteristics, while different substances may not present significantly dissimilar spectral characteristics [13], [14]. This is a great challenge for cross-scene HSIC.

Fortunately, the emergence of domain adaptation (DA) techniques can solve the problem of domain shift without requiring additional annotation costs for labeling the target domain [15], [16], [17]. Traditional DA methods such as correlation alignment (CORAL) [18] and joint distribution adaptation (JDA) [19] aim to learn transferable feature representations through distribution alignment and subspace alignment [20]. However, traditional methods can only empirically extract hand-crafted features. The advantage of a deep learning model is that it relies on a neural network to automatically extract features without human intervention, making feature extraction more comprehensive [21], [22]. The primary methods in the existing deep adaptation methods are discrepancy-based and adversarial-based ones [23].

Discrepancy-based methods aim to add adaptive layers in the network to align marginal or/and conditional distributions between domains. Adversarial-based methods learn transferable features through an adversarial game between a generator and a discriminator [24], [25]. Long et al. [26] proposed a deep adaptation network (DAN), which adds three adaptive layers and uses multikernel maximum mean discrepancy (MK-MMD) to align the marginal distribution. In addition, they designed a joint adaptation network (JAN) to learn transferable models by using joint MMD to align the joint distribution of the two domains [27].

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Considering the insufficient information contained in a single structure, Zhu et al. [28] proposed a multirepresentation adaptation network (MRAN), which aims to reduce domain discrepancy through multirepresentation alignment. Based on the concept of a subdomain, they further proposed a deep subdomain adaptation network (DSAN), which is based on local MMD and learns transferable networks by aligning the distribution of relevant subdomains [29]. For HSIC, Ma and Song [30] transferred class-level knowledge through class centroid alignment. Wang et al. [31] minimized the distribution shift between domains while maintaining the manifold structure between target samples. Huang et al. [32] proposed a spatial–spectral weighted adversarial DA (SSWADA) network, which constructs a weighted discriminator and multiclassifier structure to better align source features and target features.

Recently, self-supervised learning (SSL), including contrastive learning, has been widely used in many fields, e.g., image recognition and natural language processing [33], [34], [35]. The essence of contrastive learning lies in the comparison between samples so that the model can effectively identify similar samples and distinguish dissimilar samples [36]. In order to achieve this purpose, its guiding method is to measure the similarity between samples through a similarity function. Constructing a contrastive loss forces similar (dissimilar) samples to attract (repulse) each other, thereby learning common (discriminative) feature representations [37], [38]. To express it more accurately, in the contrastive learning framework, a sample is usually recorded as an anchor, and samples that are similar (dissimilar) to it are called positive (negative) samples.

Since contrastive learning can perform tasks well in a single domain, some work attempts to extend it to cross-domain tasks to reduce domain differences. Wang et al. [39] proposed a cross-domain contrastive learning (CDCL) framework to perform bidirectional instance-to-instance contrastive learning between two domains. Specifically, the construction of positive (negative) samples refers to samples that belong to the same (different) class as the anchor sample. Yang et al. [40] proposed a contrastive learning-assisted alignment (CLA) method, which uses instance-level contrastive loss to train target clusters and perform class conditional distribution alignment under the framework of adversarial adaptation. Target information is also used to weight the source samples, further improving the effective use of the information provided by the source data. Kim et al. [41] designed a cross-domain SSL (CDS) algorithm, which simultaneously performs instance-level SSL within and between domains to achieve instance discrimination within the domain and matching between domains.

These methods focus on contrastive learning between samples. The difference lies in the way in which positive samples are constructed, that is, by selecting samples from the same category or performing data enhancement. However, instance-level contrastive learning is easily affected by individuals. Once there are outliers or pseudo-label prediction errors, it may affect the performance of contrastive learning. Moreover, the embedding space is learned through instances, which may result in that only the local information of the instances is retained and the global semantic structure of the dataset is ignored.

Different from instance-to-instance comparison, class prototypes can be introduced into the SSL process to more robustly transfer knowledge between domains [42]. Specifically, instance-to-prototype contrastive learning is performed both within and between domains to achieve feature alignment and capture of semantic structure. Jiang et al. [43] also proposed the prototypical contrast adaptation (ProCA) method, which aligns the distribution of class centers to transfer knowledge. Singh [44] proposed a contrastive learning framework for semi-supervised DA (CLDA), which uses prototype (instance) level contrastive learning to reduce interdomain (intradomain) differences.

In HSIC problems, contrastive learning has also been applied. On a single domain, Guan and Lam [45] proposed a cross-domain contrastive learning (XDCL) framework by constructing two encoders to extract two different representations of the same sample. Domain-invariant features are captured by maximizing the consistency between the representations of positive matching pairs. In terms of few-shot learning, Liu et al. [46] proposed a refined prototypical contrastive learning network for few-shot learning (RPCL-FSL), which imposes triple constraints on the prototype to obtain more stable features. In the direction of DA, Li et al. [47] proposed a method based on supervised contrastive learning (SCLUDA), using domain similarity loss to reduce domain differences. Meanwhile, Gaussian noise and random flip operations are implemented on the samples, and instance-level contrastive learning is performed in the two domains to improve the distinguishability within the domain.

In short, in the field of computer vision, many works have applied contrastive learning to unsupervised DA problems, which can effectively achieve domain transfer and capture robust and discriminative features. However, there is currently little research on cross-scene HSIC using the idea of contrastive learning. In addition, the usual operation in contrastive learning is to regard an image as an anchor and the data-augmented sample as a positive sample. Unlike computer vision images, HSI has its own properties. In order to simultaneously preserve the spatial information and spectral information of pixels, some 3-D cubes centered on pixels are put into the network as independent samples for learning. Data enhancement methods, such as flipping and adding noise, are intuitively applied to natural images, but they may be less effective on HSI. Nonetheless, at its core, the idea of contrastive learning is very similar to the discrepancy-based method. The core is that similar samples are expected to attract each other (closer), and dissimilar samples repel each other (further distance). The difference is that in addition to considering the distribution between samples, contrastive learning also focuses on mining the information of the sample itself. Considering that the idea of contrastive learning is simple and effective for domain shift, based on the same idea, the domain invariant and compact prototype contrast adaptation (DIC-proCA) framework is proposed for cross-scene HSIC in this article.

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A prototype is a representative embedding of semantically similar instances, and the sample individual has a personality representation of the category to which it belongs. Pure instance-level or prototype-level contrastive learning may be susceptible to outliers or have difficulty in learning discriminative features. Therefore, in this article, bidirectional instance-to-prototype contrastive learning across domains is performed to learn domain-invariant features. Prototypes belonging to the same (different) category are regarded as positive (negative) samples. By strengthening the similarity (dissimilarity) between pairs of positive (negative) samples, it promotes intraclass (interclass) compactness (dispersion) in the feature space. In addition to considering domain-invariant learning between two domains, it is also important to mine the compact structure distribution within the target domain. Similarly, the data augmentation method is used to construct positive samples for anchors. In terms of the way in which negative samples are constructed, the proposed DIC-proCA is different from operations in general contrastive learning. The usual practice is that, except for the enhanced samples, all the remaining samples in a mini-batch are defaulted as negative samples, without considering potential positive samples. The proposed DIC-proCA filters out potential positive samples through predicted pseudo labels and weakens their negative impact on contrastive learning. Then, the aggregation (dispersion) between similar (dissimilar) samples within the target domain is strengthened to achieve tighter and evenly dispersed clusters.

The main contributions of the DIC-proCA are as follows.

1) A novel DA method combined with contrastive learning is proposed to effectively deal with the problem of domain shift. It mainly includes between-domain feature alignment and within-domain compact structure mining. It has excellent classification effects on three cross-scene HSIC tasks.
2) In order to promote intraclass (interclass) compactness (dispersion), prototype contrastive learning is performed between domains, and instance-aware feature learning is implemented within the target domain, thereby learning domain-invariant and discriminative features.
3) The spectral attention module (SeAM) and label smoothing (LS) strategy further improve the targeted extraction of HSI features and strengthen the stability and generalization of the model.

The remainder of the article is composed as follows. Section II describes the detailed algorithm. Section III shows the experimental results and analysis. Section IV draws the conclusion.

II. Proposed Method

The unsupervised DA technique aims to transfer a model from a source domain to a related target domain with a consistent label space, so as to complete its target task. Assume there are n_s labeled samples $\{(x_s^i, y_s^i)\}_{i=1}^{n_s}$ in the source domain $\mathcal{D}_s = (\mathcal{X}_s, \mathcal{Y}_s)$, and n_t unlabeled samples $\{x_t^j\}_{j=1}^{n_t}$ in the target domain $\mathcal{D}_t = \mathcal{X}_t$, where the two domains share the same C classes. The objective is to complete the labeling of samples in $\mathcal{D}_t$ with higher accuracy. Specifically, the trained model $h = g \circ f$ can be regarded as a combination of feature extraction network f and classifier g. For the convenience of subsequent illustration, the training samples are represented as 3-D cubes of size $p \times p \times b$. Here, p denotes the patch size, and b is the number of bands.

Fig. 1 shows the framework of the proposed DIC-proCA. It uses prototype contrastive learning for feature alignment and instance-aware feature learning in $\mathcal{D}_t$ for mining compact structures in the domain. Finally, domain transfer is achieved based on the contrastive learning strategy, and domain-invariant and discriminative features are extracted.

In addition, the specific architecture of the network f is shown in Fig. 2, which takes the Houston dataset as an example. In fact, there are many advanced feature extraction networks available for HSIs [48], [49]. Inspired by HybridSN [50], 3-D convolution with a window size of 1×1 is used to focus on extracting the spectral information of HSIs. The subsequent 2-D convolution and global average pooling operations preserve the spatial information around the pixels well. Furthermore, considering the redundancy and noise of spectral bands, a spectral attention mechanism is added to better capture spectral information. Finally, the spectral–spatial fusion features corresponding to pixels as independent samples are extracted.

A. LS Regularization

The cross-entropy (CE) loss function is commonly used in classification problems. Under the premise of unsupervised DA, learning domain-specific features of source samples is often facilitated by minimizing the CE loss. Specifically, the CE loss is calculated through the category output probability predicted by the model and the one-hot form of the real label, and then the optimal probability distribution is estimated. For the CE loss on $\mathcal{D}_s$, its general form can be expressed as

$$\mathcal{L}_s = -E_{(x_s, y_s)\in\mathcal{D}_s} \sum_{c=1}^{C} p_c \log q_c(h(x_s)) \tag{1}$$

where $q_c(\alpha) = (\exp(\alpha_c)/\sum_i \exp(\alpha_i))$ represents the probability value of the softmax output of the C-dimensional vector α for the cth class. When $\operatorname{argmax}_c q_c(h(x_s))$ corresponds to the correct class, the value of p_c is 1, and the rest are 0.

To encourage samples to form tightly and evenly separated clusters, the strategy of LS is introduced to replace the original one-hot labels [51]. Unlike 0 or 1 as the one-hot label, the label after LS is assigned a certain probability in each category. With LS, the CE loss becomes

$$\mathcal{L}_s^{ls} = -E_{(x_s, y_s)\in\mathcal{D}_s} \sum_{c=1}^{C} p_c^{ls} \log q_c(h(x_s)) \tag{2}$$

where $p_c^{ls} = (1-\varphi)p_c + \varphi/C$ is the smoothed label and φ is the hyperparameter, which generally takes a value of 0.1.

B. Spectral Attention Module

HSI has hundreds of spectral channels, and there may be spectral redundancy in addition to noise. The SeAM aims to perform weighted adjustments to the bands of input features,

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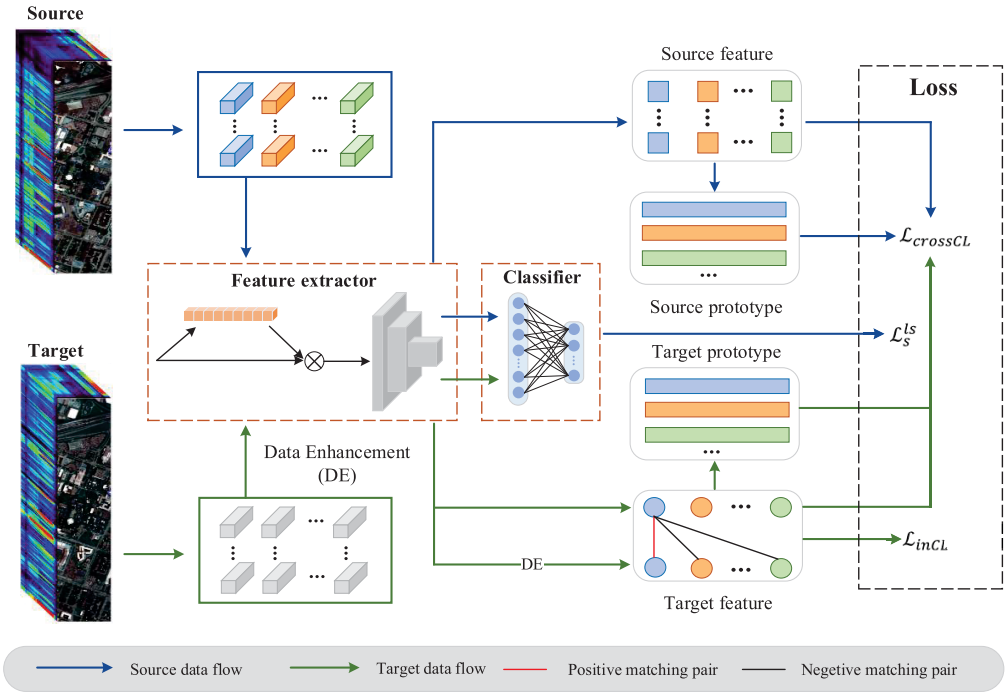


Fig. 1. Framework of the proposed DIC-proCA. Positive (negative) matching pairs represent mutual attraction (repulsion) between samples. Between the two domains, bidirectional instance-to-prototype contrastive learning is used to perform feature alignment and the domain-invariant features are learned. Within the target domain, instance-aware contrastive learning is performed between the target sample and the enhanced sample (Gaussian noise is added) to explore the compact structure in the domain, and then the discriminative features are mined.

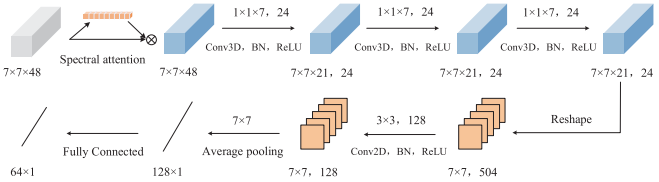


Fig. 2. Architecture of feature extraction network f.

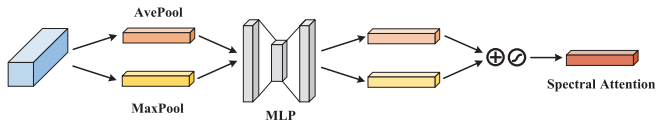


Fig. 3. Structure of SeAM.

emphasizing those spectral bands that are beneficial for feature representation [52]. Specifically, what needs to be learned is a spectral attention map $M_{\text{se}} \in \mathcal{R}^{1\times1\times b}$, which can apply weights to each band of each sample $x \in \mathcal{R}^{p\times p\times b}$. Then, the final attention calculation is as follows:

$$\bar{x} = M_{\text{se}}(x) \otimes x \tag{3}$$

where $\otimes$ denotes element-wise multiplication.

To achieve the importance weighting of spectral channels, the process of SeAM is implemented as illustrated in Fig. 3. The traditional SE mechanism only uses average pooling, while experiments in CBAM show that max-pooling features can complement the statistics of global average-pooling features [53], [54]. Their simultaneous operation allows for a more thorough encoding of statistical information. Therefore, two parallel branches are constructed here to obtain average-pooling features and max-pooling features, respectively. Global average pooling is performed on each channel of spatial size $p \times p$, as is max pooling. The kth element of z^{ave} ($z^{\max}$) is expressed as

$$z_k^{\text{ave}} = \frac{1}{p \times p} \sum_{i=1}^{p} \sum_{j=1}^{p} x_k(i, j) \tag{4}$$

$$z_k^{\max} = \max(x_k) \tag{5}$$

where $x_k(i, j)$ indicates the pixel value of the sample x at position (i, j) of the kth channel.

To reduce the amount of parameters, the two shared fully connected (FC) layers are operated on the two acquired features. Then, the two output vectors are combined by element-wise summation as follows:

$$z = \sigma\left(W_2\left(\delta\left(W_1 z^{\text{ave}}\right)\right)\right) + \sigma\left(W_2\left(\delta\left(W_1 z^{\max}\right)\right)\right) \tag{6}$$

where δ and σ refer to the ReLU and sigmoid functions, respectively, and $W_1 \in \mathcal{R}^{b'\times b}$ and $W_2 \in \mathcal{R}^{b\times b'}$ are weight matrices. In the experiment, b' is set to 16.

The final output of the SeAM is obtained by rescaling the merged features using the output activation z

$$\bar{x}_k = F_{\text{scale}}(x_k, z_k) = z_k \cdot x_k \tag{7}$$

where $\bar{x} = [\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_b]$, and $F_{\text{scale}}(x_k, z_k)$ refers to spectral-wise multiplication between the feature map $x_k \in \mathcal{R}^{p\times p\times1}$ and the scalar z_k.

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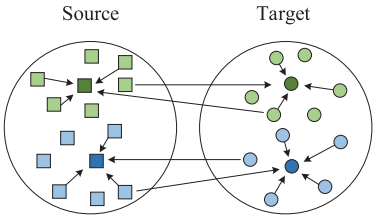


Fig. 4. Schematic of bidirectional instance-to-prototype contrastive learning. It is expected that samples or prototypes from the same category across domains can be clustered together. Here, different categories are represented in green and blue, respectively, and darker dots represent class prototypes.

C. Cross-Domain Invariant Prototype Contrastive Learning

Contrastive learning aims at clustering similar samples together and separating dissimilar samples from each other, which has achieved encouraging results in visual representation. In the case of unsupervised DA, instance-aware contrastive learning can alleviate domain shift and help learn domain-invariant features. Considering that the matching between samples is more sensitive to noisy samples, especially when there are domain differences, the category feature centroid (i.e., prototype) is introduced as it can represent the approximate distribution area of the category to which it belongs in the feature space to a certain extent. Specifically, the prototype of the cth class in $\mathcal{D}_s$ can be expressed as

$$\eta_s^c = \frac{\sum_{x_s^i \in \mathcal{X}_s} \mathbb{1}\left(y_s^i = c\right) f\left(x_s^i\right)}{\sum_{x_s^i \in \mathcal{X}_s} \mathbb{1}\left(y_s^i = c\right)}. \tag{8}$$

The prototype calculation of $\mathcal{D}_t$ can be obtained in the same way. However, since it has no real label, the pseudo label predicted by the model h is used, i.e.,

$$\hat{y}_t^j = \text{argmax}_h \, h\left(x_t^j\right). \tag{9}$$

As expected, the prototype is representative of the class, and the instance is more discriminative of its class. Domain-invariant features are learned in a contrastive manner by matching instances and prototypes across domains, as shown in Fig. 4. Specifically, for the samples in $\mathcal{D}_t$, the prototype in $\mathcal{D}_s$ that belongs to the same category is regarded as a positive sample, and the rest are regarded as negative samples. Similarly, the same matching operation is carried out for the samples in $\mathcal{D}_s$. In short, based on label information, it is hoped that samples or prototypes from the same (different) category will be clustered (separated). Therefore, given that the prototype in $\mathcal{D}_s$ is $\{\eta_s^c\}_{c=1}^C$, the feature vector in $\mathcal{D}_t$ is $v_t^j = f(x_t^j)$, and the similarity distribution vector between feature pairs is calculated as

$$P_{s\to t}^{i,j} = \frac{\exp\left(\frac{\psi\left(\eta_s^i, v_t^j\right)}{\tau}\right)}{\sum_{c=1}^C \exp\left(\frac{\psi\left(\eta_s^c, v_t^j\right)}{\tau}\right)} \tag{10}$$

where $\psi(a, b)$ is a certain metric to measure the similarity between vectors a and b, such as cosine similarity or Euclidean distance [45]. In the experiment, cosine similarity is used. Here, τ is the temperature hyperparameter. Considering that only pseudo labels are available for the target samples, the predicted labels are not accurate. Therefore, the similarity distribution vector of the sample is used to minimize the average entropy to weaken the negative impact of pseudo-labels and encourage the model to make a confident prediction. Then, the entropy of $P_{s\to t}^{i,j}$ is minimized

$$H\left(P_{s\to t}^j\right) = -\sum_{i=1}^C P_{s\to t}^{i,j} \log P_{s\to t}^{i,j}. \tag{11}$$

Similarly, the entropy $H(P_{t\to s}^i)$ between matching pairs of prototypes in $\mathcal{D}_t$ and samples in $\mathcal{D}_s$ can also be calculated. Then, the final bidirectional instance-to-prototype contrastive loss is

$$\mathcal{L}_{\text{crossCL}} = \frac{1}{n_t}\sum_{j=1}^{n_t} H\left(P_{s\to t}^j\right) + \frac{1}{n_s}\sum_{i=1}^{n_s} H\left(P_{t\to s}^i\right). \tag{12}$$

D. Within Domain Compact Instance-Aware Feature Learning

Domain-invariant representation features can be mined through instance-to-prototype contrastive learning between domains. However, due to the existence of domain differences and the use of pseudo labels in $\mathcal{D}_t$, the learned target distribution may not be compact enough. In order to increase the semantic representation of target features and the discriminative ability of the model, data augmentation technology is applied to target samples. Considering the characteristics of HSIs, Gaussian noise is added. Naturally, instance-level contrastive learning is implemented in $\mathcal{D}_t$ to achieve stable and accurate target clustering.

The core of contrastive learning lies in constructing positive (negative) samples corresponding to the anchor samples, and the construction of appropriate positive (negative) matching pairs is conducive to the success of contrastive learning. In $\mathcal{D}_t$, the sample after data enhancement is recorded as the positive sample $\tilde{\mathcal{X}}_t = \{\tilde{x}_t^1, \tilde{x}_t^2, \ldots, \tilde{x}_t^{n_t}\}$ corresponding to the anchor sample $\mathcal{X}_t = \{x_t^1, x_t^2, \ldots, x_t^{n_t}\}$. Considering the general practice of defaulting the remaining samples to be negative samples, there may be many potential positive samples, which have a negative impact on contrastive learning. Therefore, using predicted pseudo labels, samples whose categories are different from the anchor samples in $\mathcal{X}_t \cup \tilde{\mathcal{X}}_t$ are screened out as negative samples. Then, the instance-aware contrastive loss of x_t^j and $\tilde{x}_t^j$ can be expressed as

$$\begin{aligned}&\mathcal{L}_{\text{inCL}}\left(x_t^j, \tilde{x}_t^j\right) \\ &= -\log \frac{\exp\left(\frac{\psi\left(v_t^j, \tilde{v}_t^j\right)}{\tau}\right)}{\exp\left(\frac{\psi\left(v_t^j, \tilde{v}_t^j\right)}{\tau}\right) + \sum_{v_t^k \in A(k)} \exp\left(\frac{\psi\left(v_t^j, v_t^k\right)}{\tau}\right)}\end{aligned} \tag{13}$$

where $v_t^j = f(x_t^j)$ and $\tilde{v}_t^j = f(\tilde{x}_t^j)$ represent the learned features, and $A(k)$ represents the set $\{f(x_t^k) | x_t^k \in \mathcal{X}_t \cup \tilde{\mathcal{X}}_t, \hat{y}_t^k \neq \hat{y}_t^j\}$. The above formula is based on x_t^j as the anchor. Symmetrically, $\mathcal{L}_{\text{inCL}}(x_t^j, \tilde{x}_t^j) \neq \mathcal{L}_{\text{inCL}}(\tilde{x}_t^j, x_t^j)$ when $\tilde{x}_t^j$ is

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denoted as an anchor. Therefore, the final in-domain instance-aware contrastive loss is

$$\mathcal{L}_{\text{inCL}} = \frac{1}{2n_t}\sum_{j=1}^{n_t}\left(\mathcal{L}_{\text{inCL}}\left(x_t^j, \tilde{x}_t^j\right) + \mathcal{L}_{\text{inCL}}\left(\tilde{x}_t^j, x_t^j\right)\right). \quad (14)$$

E. Total Loss Function

In order to achieve model transfer between two domains with different distributions, prototype contrast adaptation is applied to achieve this purpose. Cross-domain prototype contrastive learning is used to mine domain-invariant representations. Meanwhile, instance-level comparison within $\mathcal{D}_t$ promotes compact class clustering and improves the discriminability of samples. The LS-assisted CE loss ensures the classification accuracy of $\mathcal{D}_s$ and improves the stability of the model. Therefore, the final total loss function is

$$\mathcal{L} = \mathcal{L}_s^{ls} + \lambda\mathcal{L}_{\text{crossCL}} + \mu\mathcal{L}_{\text{inCL}} \quad (15)$$

where λ and μ are tradeoff parameters. The pseudo-code of DIC-proCA is shown in Algorithm 1.

Algorithm 1 DIC-proCA

Input: The source sample set $\{x_s^i, y_s^i\}_{i=1}^{n_s}$, the target sample set $\{x_t^j\}_{j=1}^{n_t}$, the total epoch number E, the iterations per epoch R, the trade-off parameters λ and μ.
Output: Predicted labels for $\mathcal{D}_t$.
Begin

Randomly initialize model h.
for $e = 1$ to E **do**
  for $r = 1$ to R **do**
    Perform data enhancement on the original target sample to obtain $\tilde{\mathcal{X}}_t$.
    Learn the feature representation $f(x)$ and obtain the target pseudo label by Eq. (9).
    Calculate the category prototype of $\mathcal{D}_s$ according to Eq. (8), and similarly calculate the category prototype of $\mathcal{D}_t$ based on Eq. (9).
    Obtain the prototype contrastive loss $\mathcal{L}_{crossCL}$ between two domains by Eq. (12).
    Obtain the instance-aware contrastive loss $\mathcal{L}_{inCL}$ within $\mathcal{D}_t$ by Eq. (14).
    Back propagate and update the model by Eq. (15).
  end for
end for
Get the target prediction label $\mathcal{Y}_t$.

III. Experimental Results

A. Data Description

Three HSI datasets are used to verify the effectiveness of the DIC-proCA, namely, the Houston dataset, the Hangzhou–Shanghai dataset, and the Huanghekou dataset.

1) Houston: This dataset was acquired over the University of Houston campus and surrounding areas with different sensors in 2013 [55] and 2018 [56]. The Houston 2013 (Houston 2018) dataset contains 349 × 1905 (209 × 955) pixels and 144 (48) spectral bands. In the experiment, the corresponding 48 spectral bands and seven common categories are used, and the specific category information is shown in Table I.

TABLE I
Land Cover Class and Numbers on the Houston Dataset

Class		Number	
No.	Name	Houston 2013 (source)	Houston 2018 (target)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
Total		2530	53200

TABLE II
Land Cover Class and Numbers on the HS Dataset

Class		Number	
No.	Name	Hangzhou (source)	Shanghai (target)
1	Water	18043	123123
2	Land/building	77450	161689
3	Plant	40207	83188
Total		135700	368000

2) Hangzhou–Shanghai (HS): The dataset was acquired by the EO-1 Hyperion hyperspectral sensor, which includes 198 spectral bands after processing [57]. The image sizes of Hangzhou and Shanghai are 590 × 230 and 1600 × 230. The three shared categories are water, land/buildings, and plants. The details are shown in Table II.

3) Huanghekou (HHK): The two datasets were imaged over the Huanghekou area at different times by the ZY1-02D satellite, on June 28, 2020 and September 29, 2021 [58]. The image sizes are 1147 × 1600 and 1050 × 1219, respectively. Eight common categories are used for classification, and detailed land cover categories are shown in Table III.

The pseudo-color images and ground-truth maps are shown in Figs. 5–7. For clearer representation, the background color of the Houston task is black, and that of the HHK task is white.

B. Experiment Setup

The proposed DIC-proCA algorithm is compared with some other DA algorithms, including KNN (no adaptation), CORAL [18], JDA [19], DAN [26], DSAN [29], MRAN [28], CDCL [39], CLDA [44], and SCLUDA [47].

The subspace dimension of JDA is $k = 20$, and traditional DA methods all use a 1-NN classifier. In the deep DA algorithm, regarding the patch size, the Houston task and HHK task are 7 × 7, and the HS task is 3 × 3. On all tasks, the number of iterations is 100, the batch size is set to 128, and the initial learning rates are all selected

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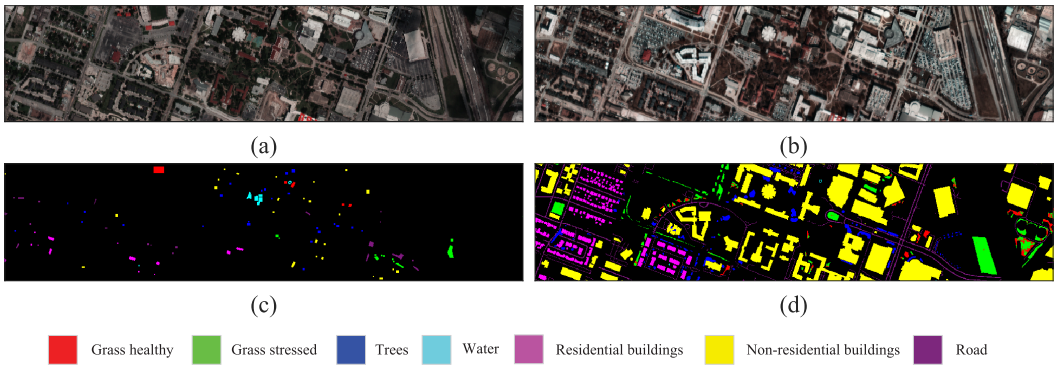


Fig. 5. Houston image. Pseudo color image of (a) Houston 2013 and (b) Houston 2018. Ground-truth map of (c) Houston 2013 and (d) Houston 2018.

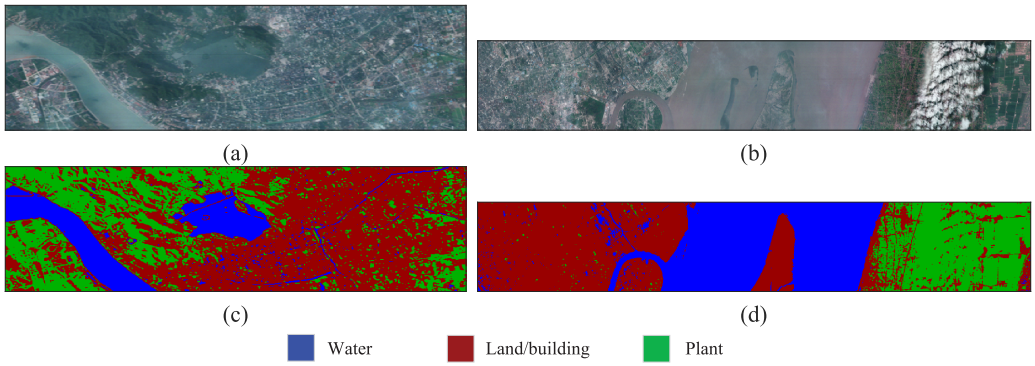


Fig. 6. HS image. Pseudo-color image of (a) Hangzhou and (b) Shanghai. Ground-truth map of (c) Hangzhou and (d) Shanghai.

TABLE III
LAND COVER CLASS AND NUMBERS ON THE HHK DATASET

Class		Number	
No.	Name	HHK20200628 (source)	HHK20210929 (target)
1	Reed	310	1057
2	Salt Flat Filtration Pond	247	512
3	Salt Flat Evaporation Pond	300	260
4	Salt Flat	306	412
5	Suaeda	218	418
6	River	584	941
7	Sea	4694	10239
8	Tide Ditch	67	248
Total		6726	14087

from $\{1e^{-5}, 1e^{-4}, 1e^{-3}, 1e^{-2}, 1e^{-1}\}$. The CDCL, CLDA and DIC-proCA algorithms related to contrastive learning have a temperature parameter $\tau = 0.5$ [59].

For quantitative assessment, overall accuracy (OA) and kappa coefficient (κ) are presented in the experiments.

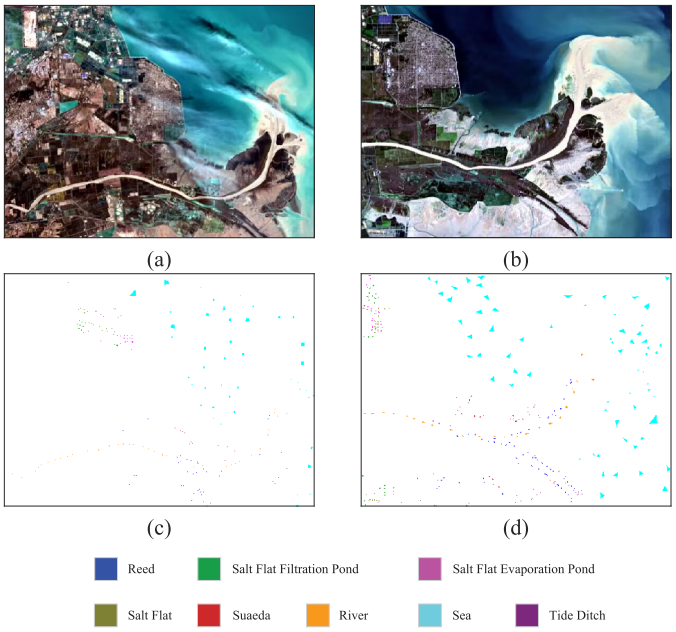


Fig. 7. HHK image. Pseudo color image of (a) HHK20200628 and (b) HHK20210929. Ground-truth map of (c) HHK20200628 and (d) HHK20210929.

C. Experiments

1) Experiments on the Houston Task: The Houston task is more challenging because it is collected by different sensors.

Table IV displays the classification results of different DA algorithms. When no DA strategy is used and only 1-NN classification is performed on the target domain, an accuracy of 69.49% can be achieved. Traditional and general deep DA methods are even inferior to KNN classification. This may be because simply reducing the distribution difference between domains cannot extract good feature representation

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TABLE IV
CLASSIFICATION ACCURACIES ON THE HOUSTON DATASET

Class	KNN	CORAL	JDA	DAN	DSAN	MRAN	CDCL	CLDA	SCLUDA	DIC-proCA
1	78.94	**98.60**	81.82	78.17	83.06	76.40	60.62	73.47	90.98	75.35
2	**94.82**	46.32	80.36	68.31	67.96	68.21	71.58	78.97	76.92	83.95
3	66.52	64.46	65.80	56.83	**75.00**	72.04	54.32	69.27	65.37	70.52
4	**100**	**100**	**100**	96.36	99.35	94.89	98.86	90.91	**100**	80.00
5	64.00	73.48	60.50	56.72	65.53	67.51	77.11	68.90	**84.37**	67.03
6	68.10	74.90	59.86	71.30	66.25	65.41	70.51	76.43	72.16	**86.86**
7	60.91	50.73	**72.07**	55.02	60.49	66.75	61.72	33.56	48.00	62.62
OA(%)	69.49	69.31	64.15	67.05	66.54	66.67	69.14	70.34	71.07	**80.56**
κ	0.539	0.522	0.487	0.501	0.516	0.516	0.531	0.528	0.565	**0.680**

TABLE V
CLASSIFICATION ACCURACIES ON THE HS DATASET

Class	KNN	CORAL	JDA	DAN	DSAN	MRAN	CDCL	CLDA	SCLUDA	DIC-proCA
1	93.35	88.20	91.66	91.96	90.41	92.78	92.96	**93.60**	89.07	93.37
2	41.51	66.20	72.88	67.58	69.72	66.96	80.04	84.05	81.20	**94.35**
3	**100**	95.94	91.27	**100**	99.91	99.03	99.97	99.79	99.72	96.17
OA(%)	72.07	80.28	83.32	83.07	83.47	82.85	88.87	90.80	88.02	**94.43**
κ	0.599	0.706	0.748	0.748	0.753	0.745	0.831	0.860	0.818	**0.914**

and cannot perform effective model transfer. In addition, the feature extraction network based on ResNet-50 may not be suitable for HSI and cannot properly represent spatial-spectral features. Both CDCL and CLDA methods are DA strategies based on contrastive learning, which use the idea of comparison between samples to learn transferable features. Although ResNet-50 is still used as the basic network, the performance is better than the discrepancy-based DA algorithm, about 3%–4% higher. Finally, the proposed DIC-proCA specifically extracts the features of HSI and learns domain-invariant and compact features based on contrastive learning. The performance is far better than other algorithms, reaching 80.56%.

2) Experiments on the HS Task: The HS task is relatively simple, with only three distinct categories that are easy to distinguish.

Table V shows the experimental results of different DA algorithms. When only using the 1-NN classifier, the classification result can reach 72.07%. The CORAL and JDA methods use linear transformation to align the statistics and joint distributions of the two domains, improving performance by 8% and 11%, respectively. The DAN, DSAN, and MRAN methods use deep neural networks for nonlinear learning and use adaptive layers to align marginal or/and conditional distributions, and the classification performance reaches about 83%. As for the CDCL and CLDA algorithms related to contrastive learning, the former focuses on contrastive learning between samples, and the latter implements contrastive learning at the prototype level and sample level. Compared with the general deep DA algorithm, their classification results of the Shanghai scene are nearly 5% higher, reaching 88.87% and 90.80%, respectively. The proposed DIC-proCA uses 3-D, 2-D convolution, and spectral attention to extract more discriminative feature representations, and also utilizes contrastive learning to transfer features. Then, in terms of classification results, it is 5.56% and 3.63% higher than CDCL and CLDA, respectively. In addition to outperforming in OA, DIC-proCA also produces a very high value of kappa, and it achieves classification accuracy of over 93% in all three categories.

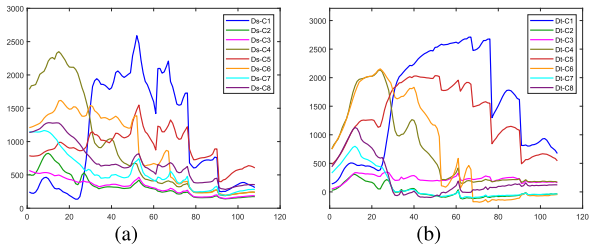


Fig. 8. Mean spectrum of all classes on the HHK task. (a) Source domain. (b) Target domain.

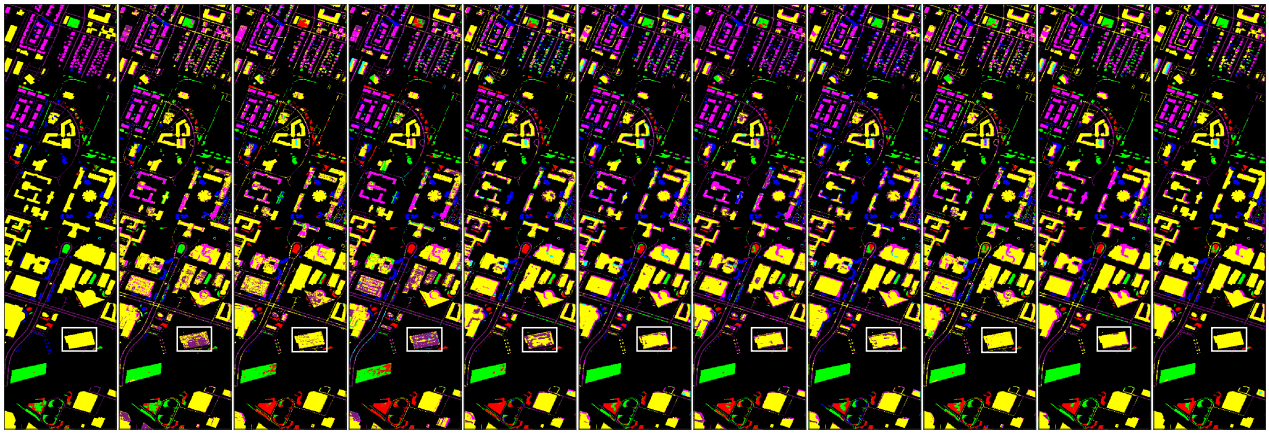
3) Experiments on the HHK Task: Due to the impact of collection time, environment, and changes in ground objects, the classification of this task is very difficult.

Table VI lists the classification results of different algorithms. When DA is not used, the result is very unsatisfactory, only 14.44%. When traditional linear transformation is used for alignment, the performance is significantly improved, and the results obtained by the CORAL and JDA methods are 41.35% and 52.66%, respectively. When using deep learning methods to automatically extract features and reduce domain discrepancy based on MMD and its variants, the OA on the Huanghekou scene can approach about 80%. The CDCL method performs contrastive learning between cross-domain samples. Although it does not directly reduce domain differences, it can also better learn transferable features, achieving a classification result

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TABLE VI
CLASSIFICATION ACCURACIES ON THE HHK DATASET

Class	KNN	CORAL	JDA	DAN	DSAN	MRAN	CDCL	CLDA	SCLUDA	DIC-proCA
1	**97.35**	3.97	0	46.17	63.47	70.06	94.31	95.27	68.97	94.69
2	1.56	3.91	14.26	0	0.16	0.83	1.14	0	**34.18**	12.50
3	12.69	20.00	5.38	0.26	1.22	6.15	4.49	0	51.15	**99.23**
4	96.60	43.20	77.67	5.34	1.42	16.75	52.87	0.24	**98.54**	89.32
5	**55.26**	0	0	24.64	14.71	24.34	24.92	11.48	6.22	22.04
6	30.50	**100**	0.43	92.88	96.58	82.33	52.05	97.98	99.79	**100**
7	0.02	44.73	68.25	97.22	96.14	94.59	93.29	**99.66**	95.91	96.58
8	18.55	4.84	7.66	0	0	0	4.84	0	**67.34**	5.24
OA(%)	14.44	41.35	52.66	81.22	81.60	80.87	80.85	86.48	87.99	**89.63**
κ	0.116	0.064	0.224	0.495	0.525	0.517	0.557	0.648	0.745	**0.778**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

Fig. 9. Houston task. (a) Ground truth, (b) KNN (69.49%), (c) CORAL (69.31%), (d) JDA (64.15%), (e) DAN (67.05%), (f) DSAN (66.54%), (g) MRAN (66.67%), (h) CDCL (69.14%), (i) CLDA (70.34%), (j) SCLUDA (71.07%), and (k) DIC-proCA (80.56%).

of 80.85%. The CLDA method can achieve 86.48% in OA, but it is almost completely wrong in categories 2, 3, 4, and 8. It may be that implementing prototype-to-prototype contrastive learning between domains fails to learn more refined features. The proposed DIC-proCA performs instance-to-prototype contrastive learning across domains, taking into account both the discriminability of the sample and the class representativeness of the prototype. The mining of compact discriminative features in $\mathcal{D}_t$ is also performed. Finally, a classification performance of 89.63% is achieved on the Huanghekou scene. Meanwhile, the kappa coefficient far exceeds other comparison algorithms. In addition, the mean spectral curves for all categories of the two domains on this task are shown in Fig. 8. It is obvious that the spectral curves of some categories are very similar, such as categories 2, 7, and 8, which brings great difficulty to the distinction between these land cover categories. It is worth mentioning that the traditional DA algorithms and the simple deep DA algorithms only perform marginal or conditional distribution alignment, without carefully paying attention to the fine-grained learning of the sample itself. Therefore, several categories are completely misclassified, and the result is 0. However, the CDCL, SCLUDA, and DIC-proCA algorithms use contrastive learning between samples, which can well distinguish the differences between samples of different categories, and correctly classify some samples in each category.

Figs. 9–11 also show classification maps. It can be clearly observed that the proposed DIC-proCA is more consistent with the ground truth, especially the area marked by the rectangular box.

D. Ablation Analysis

To validate each module of the proposed DIC-proCA, an ablation experiment is conducted on the Houston task as shown in Table VII.

Experiments 1–7 all use the standard CE loss $\mathcal{L}_s$ to ensure the classification accuracy of $\mathcal{D}_s$. Among them, without any DA method, just using $\mathcal{L}_s$ to train the model and directly label the target sample, the result is only 66.44%. On this basis, experiment 2 adds a cross-domain prototype contrastive loss, and the classification performance is as good as 75.49%. It appears that contrastive learning between two domains can mitigate the negative impact of domain differences on model transfer. Similarly, experiment 3 adds instance-level contrastive learning in $\mathcal{D}_t$, and its classification performance is even 0.68% higher than experiment 2. This indicates that fine-grained learning of samples within a domain in cross-scene tasks may be more important than

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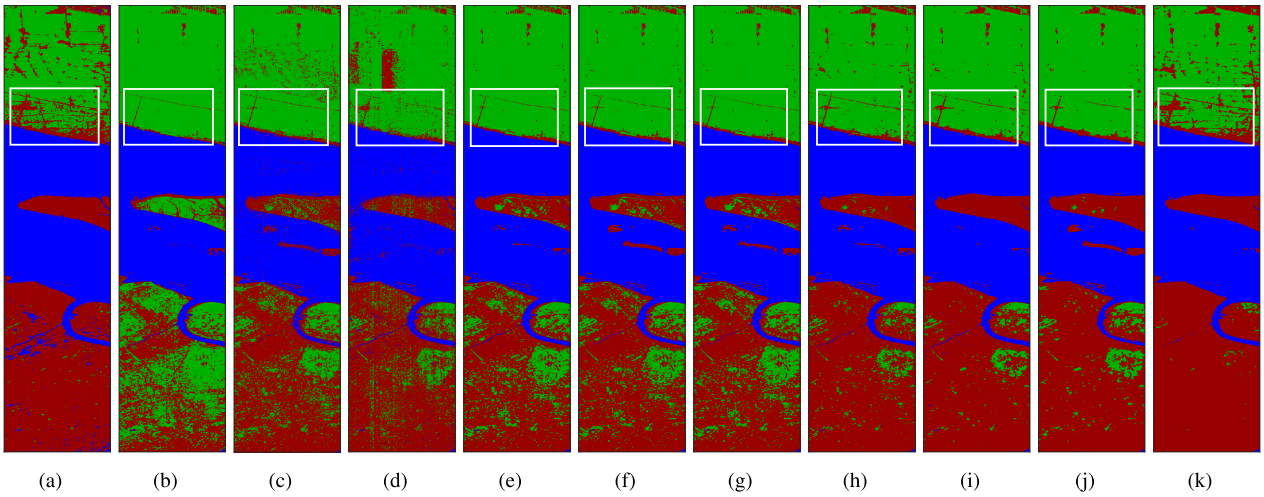


Fig. 10. HS task. (a) Ground truth, (b) KNN (72.07%), (c) CORAL (80.28%), (d) JDA (83.32%), (e) DAN (83.07%), (f) DSAN (83.47%), (g) MRAN (82.85%), (h) CDCL (88.87%), (i) CLDA (90.80%), (j) SCLUDA (88.02%), and (k) DIC-proCA (94.43%).

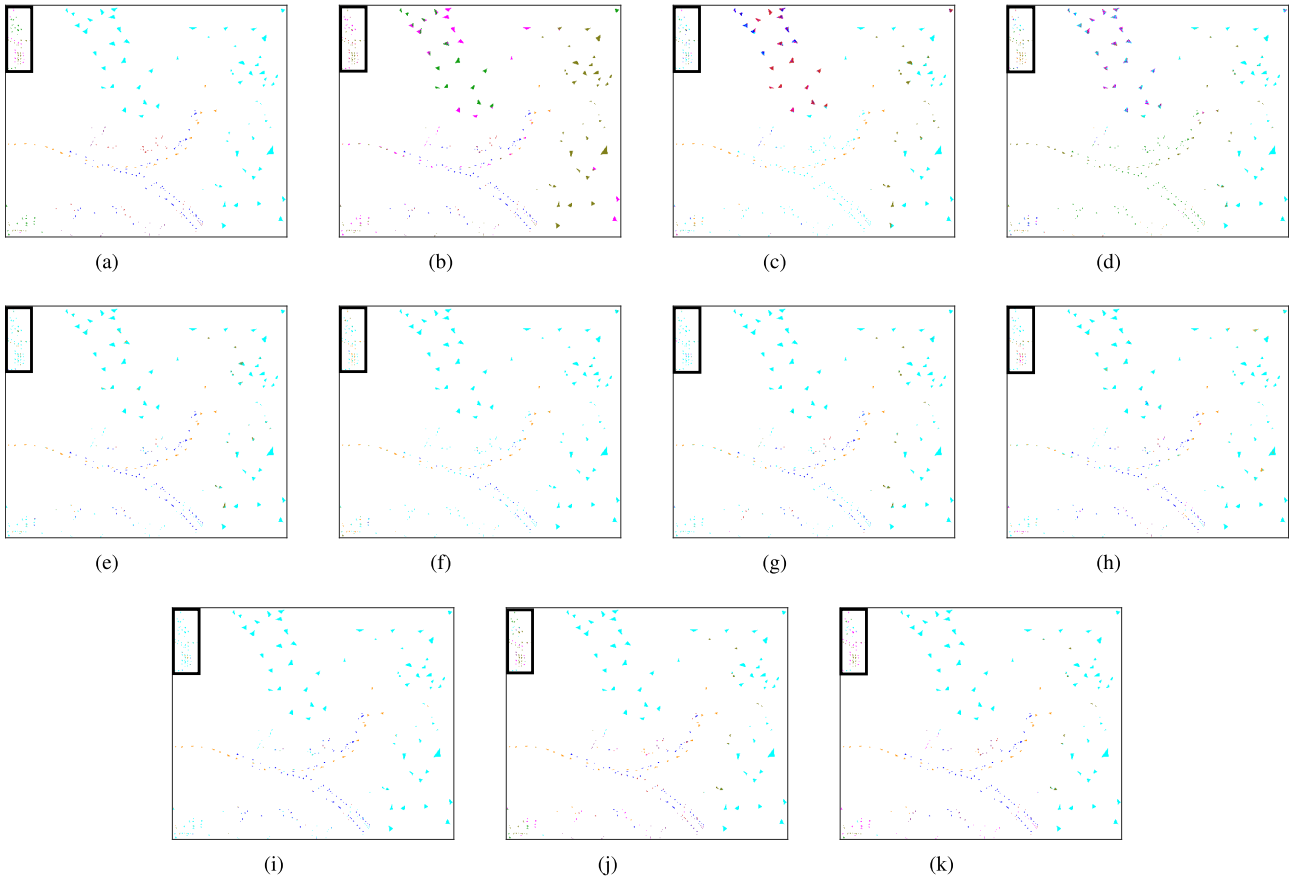


Fig. 11. HHK task. (a) Ground truth, (b) KNN (14.44%), (c) CORAL (41.35%), (d) JDA (52.66%), (e) DAN (81.22%), (f) DSAN (81.60%), (g) MRAN (80.87%), (h) CDCL (80.85%), (i) CLDA (86.48%), (j) SCLUDA (87.99%), and (k) DIC-proCA (89.63%).

reducing domain shifts. The superposition of the two losses between interdomain and intradomain not only facilitates the representation of domain-invariant features but also learns fine-grained compact features. Therefore, experiment 4 achieves an accuracy of 78.15%.

In order to examine whether the attention mechanism plays a role in the algorithm, experiments 5 and 6 add SeAM on the basis of experiments 2 and 3, respectively. Considering that HSIs have a lot of spectral bands, SeAM can weigh the bands, so that the model can focus on important bands when learning. In terms of classification performance, it is improved by 0.73% and 0.88%, respectively. In experiment 7, when the two contrastive losses simultaneously act, SeAM with the consideration of band importance can achieve an accuracy of 79.08%, which is an increase of 0.93% compared with experiment 4.

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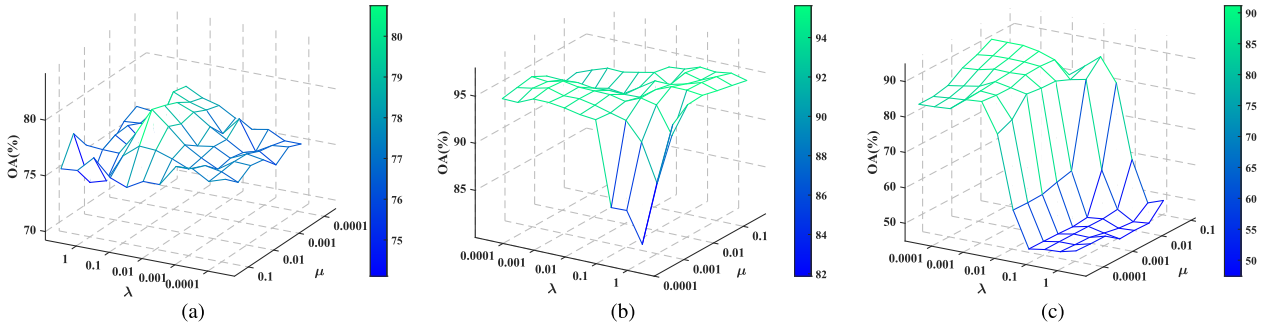


Fig. 12. OA of DIC-proCA versus parameters λ and μ on (a) Houston task, (b) HS task, and (c) HHK task.

TABLE VII
ABLATION STUDY OF DIC-PROCA FOR HOUSTON TASK

	$\mathcal{L}_s$	LS	$\mathcal{L}_{crossCL}$	$\mathcal{L}_{inCL}$	SeAM	OA(%)
1	✓					66.44
2	✓		✓			75.49
3	✓			✓		76.17
4	✓		✓	✓		78.15
5	✓		✓		✓	76.22
6	✓			✓	✓	77.05
7	✓		✓	✓	✓	79.08
DIC-proCA	✓	✓	✓	✓	✓	**80.56**

Finally, the proposed DIC-proCA adds LS regularization to standard $\mathcal{L}_s$, forcing the clusters to be compact and evenly separated. The spectral features are weighted through SeAM, and two losses are used to promote feature alignment and discriminative learning in a contrastive manner. Furthermore, in the feature space, intraclass compactness and interclass dispersion are promoted. Therefore, the combination of all modules enables the task to be better accomplished, achieving a classification result of 80.56%.

E. Parameter Analysis

In the optimization function of the proposed DIC-proCA, two tradeoff parameters λ and μ are included. Their function is to control the contribution of CDCL and in-domain compact feature learning to cross-scene classification. In order to set appropriate parameters, we change parameters λ and μ in [0.0001, 1] and then observe the fluctuations in classification performance caused by different combinations on the three tasks. In Fig. 12(a), it can be seen that when $\lambda = 0.1$, $\mu = 0.01$, the performance on the Houston task is the best, and the fluctuations nearby are also relatively gentle. In order to keep the parameters as consistent as possible, the suboptimal classification result is selected according to Fig. 12(b). In the HS scenario, λ and μ still take the values 0.1 and 0.01. On the HHK task, ablation experiments indicate that feature learning within the domain may be more important. Therefore, the corresponding values of parameters λ and μ on this task are 0.001 and 0.01, respectively.

TABLE VIII
ANALYSIS OF PATCH SIZE ON THREE TASKS

	3×3	5×5	7×7	9×9
Houston	74.37%	78.58%	**80.56%**	78.64%
HS	**94.43%**	92.87%	91.96%	90.83%
HHK	87.30%	88.96%	**89.63%**	83.69%

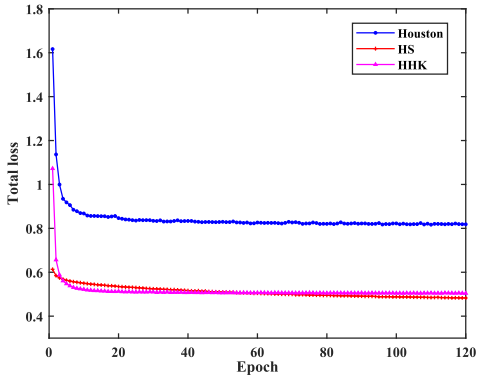


Fig. 13. Convergence analysis of DIC-proCA on three tasks.

Patch size defines the range of spatial information to be extracted from HSIs, and choosing an appropriate patch size is important for HSIC. In order to study its impact, different sizes are selected from 3 × 3, 5 × 5, 7 × 7, and 9 × 9 for experiments. As shown in Table VIII, on the Houston and HHK tasks, when the patch size is 7 × 7, the classification results reach the peak. For the HS classification task, the optimal patch size is 3 × 3.

In addition, convergence analysis is performed on three tasks to determine the number of training iterations. It can be found from Fig. 13 that as the model continues to train, the total loss gradually stabilizes. Taking into account selecting an appropriate number of training iterations to reduce computational overhead, the number of training epochs is uniformly set to 100.

F. Feature Visualization

To verify the degree of aggregation of samples by the proposed DIC-proCA algorithm and whether it can effec-

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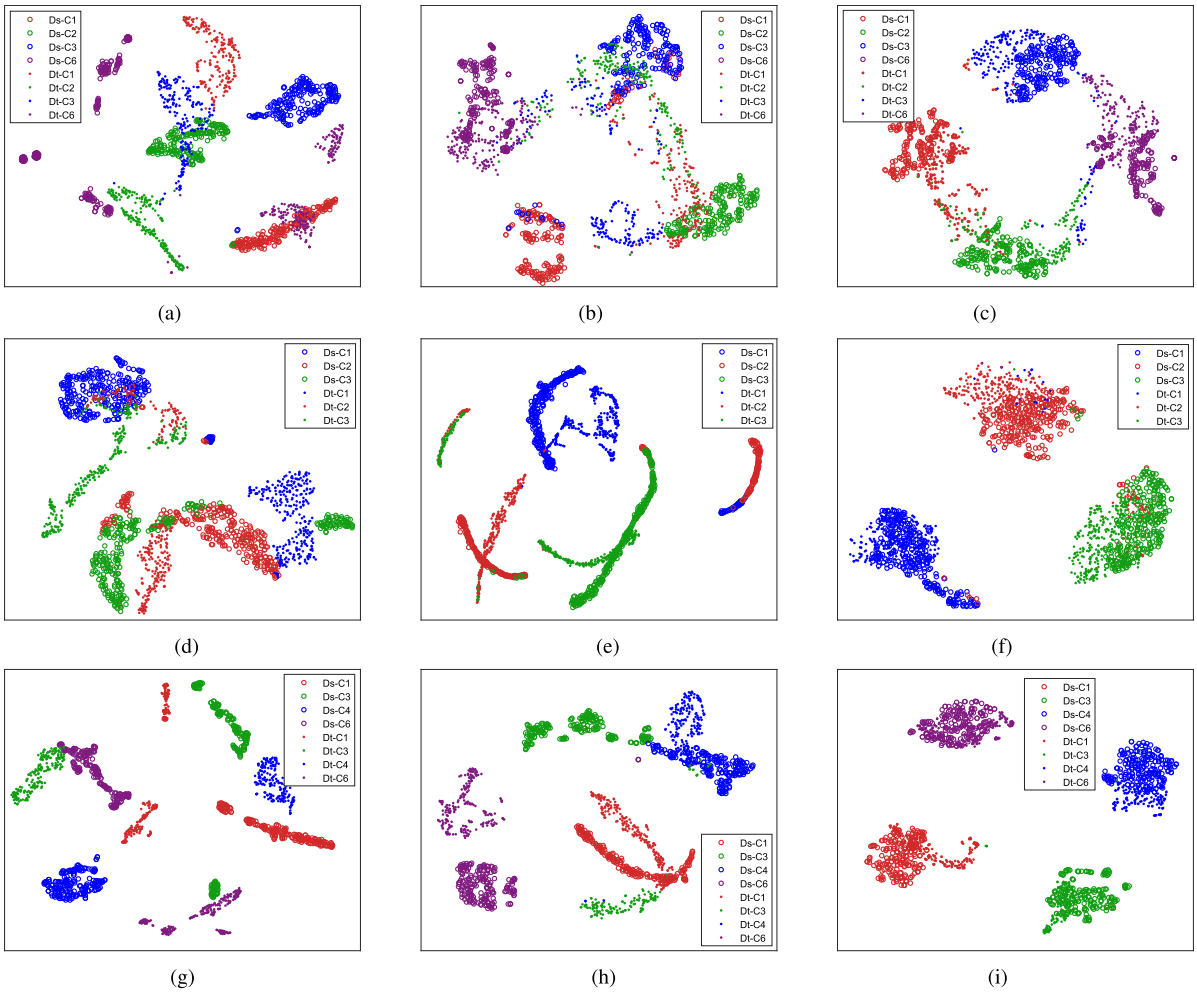


Fig. 14. tSNE feature visualization on three tasks. Circles (points) represent source (target) samples. Different colors represent different categories. The visualization of the original feature is shown in (a), (d), and (g). The visualization of features learned by the CDCL algorithm is shown in (b), (e), and (h). The visualization of features learned by our DIC-proCA algorithm is shown in (c), (f), and (i). (a) Original for Houston task. (b) CDCL for Houston task. (c) DIC-proCA for Houston task. (d) Original for HS task. (e) CDCL for HS task. (f) DIC-proCA for HS task. (g) Original for HHK task. (h) CDCL for HHK task. (i) DIC-proCA for HHK task.

tively reduce domain discrepancy, tSNE is used to visualize the distribution of features on the three tasks. For better comparison, the original data, and features after CDCL and DIC-proCA methods are shown in Fig. 14. It simultaneously shows the distribution of samples in the two domains. From Fig. 14(a), (d), and (g), it is obvious that the original samples from the two domains are widely separated, and the distribution is completely inconsistent. Some categories are even divided into several clusters, such as the purple source samples in the Houston task. After contacting the two domains through the CDCL, it is clearly shown in Fig. 14(b), (e), and (h) that the distance between source and target samples of some categories is much closer. It can be seen that the samples in the two domains are divided into several clusters according to categories. For the proposed DIC-proCA, it can be more clearly observed from Fig. 14(c), (f), and (i) that the clusters are densely and evenly distributed. This indicates that contrastive learning-based methods can effectively reduce domain differences and learn domain-invariant and discriminative features.

TABLE IX
ONE EPOCH TRAINING TIME (IN SECONDS) FOR DIFFERENT METHODS

methods	DAN	DSAN	MRAN	CDCL	CLDA	SCLUDA	DIC-proCA
time(s)	4.567	4.691	4.769	3.181	4.085	4.012	**2.726**

G. Analysis of the Computational Complexity

Time complexity is also an important measurement criterion in algorithm comparisons. All experiments are performed with a 2.7 GHz Intel Xeon Gold 6258R CPU and two NVIDIA GeForce RTX 2080 Ti GPUs. As shown in Table IX, the training time of different algorithms in one epoch is displayed on the Houston task. The proposed DIC-proCA algorithm offers the best performance but costs the least amount of training time.

IV. CONCLUSION

In this work, a novel prototype contrast adaptive framework is proposed to handle cross-scene HSIC. The instance-aware contrastive learning is prone to be affected by abnormal

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samples, it is difficult to learn the discriminability of the sample itself for prototype-aware contrastive learning, not even mention that there are differences between the two domains. Therefore, prototypes are introduced into the model, and bidirectional instance-to-prototype contrastive learning between domains connects the two domains to learn domain-invariant feature representations. Meanwhile, discriminative compact features are learned for original samples and data-augmented samples in $\mathcal{D}_t$ in a contrastive manner. In addition, clusters are forced to separate tightly and evenly through LS, which also results in a more robust and generalized model. The experimental results on three tasks demonstrate the effectiveness of the proposed DIC-proCA.

During prototype contrastive learning, the class prototype is calculated from samples belonging to the same class in a batch. However, due to the randomness of mini-batch training, the samples of each category put into training are not uniform, so the prototype may not be too stable. In future work, calibration of the prototype will be considered to learn a more stable and accurate representation of the prototype and improve the quality of contrastive learning.

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